# Linear Algebra MAT 226

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## Homework #6

### Exercise 2.2

8) $A = PBP^{-1}$

$P$ is invertible

(Matrices are not commutative)

$P^{-1}A = P^{-1}(PBP^{-1})$

$P^{-1}A = BP^{-1}$

$(P^{-1}A)P = BP^{-1}.P$

$P^{-1}AP = B.I$

$P^{-1}AP = B$

$B = P^{-1}AP$

10) (a) False

the elementary row operations that reduce $A$ to $I_n$ would actually reduce $A^{-1}$ to $A^{-1}.A^{-1}$ (and not $I_n$).

(b) True

by definition $A \cdot A^{-1} = I$ & $A^{-1}A = I$

so $(A^{-1})^{-1} = A$

(c) False

the inverse of product is not in the same order
$(AB)^{-1} = B^{-1} \cdot A^{-1}$

(d) True

If $A\vec{x} = e_j$

then you can find a solution for every column of the identity matrix. $X = [\vec{x_1} \cdots \vec{x_n}]$

$$AX = I$$

This guarantees A is invertible.

(e) True

if $A \sim I$ it means the matrix has pivot position in every row & every column. This means no free variables and a full span making the matrix invertible.

22)

If nxn matrix A is invertible, then for any vector $\vec{b}$ in $\mathbb{R}^n$ $A\vec{x} = \vec{b}$ has a unique solution, where

$$\vec{x} = A^{-1}\vec{b}$$

because there is a valid $\vec{x}$ for every $\vec{b}$ in $\mathbb{R}^n$, $A\vec{x} = \vec{b}$ is consistent for all $\vec{b}$ in codomain.

This means that $A\vec{x} = \vec{b}$ having a solution for each $\vec{b}$ in $\mathbb{R}^n$ is the same as saying that the columns of A span $\mathbb{R}^n$. Since the solution is guaranteed to exist, it means that every vector

in $\mathbb{R}^n$ can be formed by a linear combination of the columns of A, meaning the columns span $\mathbb{R}^n$.

26) show inverse formula works when $ad-bc \neq 0$

$$AA^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \left( \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \right)$$

$$= \frac{1}{ad-bc} \times \left( \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \right)$$

$$= \frac{1}{ad-bc} \times \left( \begin{bmatrix} (ad - bc) & (-ab + ba) \\ (cd - dc) & (-cb + da) \end{bmatrix} \right)$$

$$= \frac{1}{ad-bc} \times \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$$

$$= \begin{bmatrix} \frac{ad-bc}{ad-bc} & 0 \\ 0 & \frac{ad-bc}{ad-bc} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I_2$$

## Exercise 2.3

(2) $\begin{bmatrix} -4 & 2 \\ 6 & -3 \end{bmatrix}$

$$ad - bc = (-4)(-3) - (2)(6)$$
$$= 0$$

The determinant is zero so matrix is not invertible.

(4) $\begin{bmatrix} -5 & 1 & 4 \\ 0 & 0 & 0 \\ 1 & 4 & 9 \end{bmatrix}$

This Matrix can never have a pivot position in Row 2, so its columns cannot span $\mathbb{R}^3$.

If a square matrix doesnot have a pivot in every row, it is not invertible.

12) (a) True

$AD = I$ the $DA = I$
inverses are unique so any $X$ such that $X = IA^{-1}$
or $X = A^{-1}I$
is $D$

(b) False
this is not true if $A$ is not invertible.

(c) True

For $n \times n$ matrices, linear independence means $n$ pivot columns ensuring a pivot in every row hence columns span $\mathbb{R}^n$.

(d) False
A transformation that is onto is also one-to-one.

(e) False

consistency means that atleast one solution exists for $\vec{b}$. It does not guarantee that the solution is unique.

14)

For an nxn matrix to be invertible, it must be row-equivalent to the identity matrix, and it needs to have n pivot positions.

A lower triangle matrix has zeros everywhere above the main diagonal, the diagonal entries are the only way to fit all n pivots. Hence, all entries in main diagonal must be non-zero.

16) If nxn matrix is invertible then its transpose is also invertible. Any invertible matrix must have linearly independent columns hence the columns of $A^T$ are linearly independent

18) A square matrix with 2 identicle rows is not invertible because elementary row operations would produce a row of zeros. Thes matrix would not have a pivot in every column and cannot be reduced to the identity matrix and is therefore not invertible.

20) No, it is not possible.

$A\vec{x} = \vec{b}$ is consistent for every $\vec{b}$ in $\mathbb{R}^5$.

The columns of A span $\mathbb{R}^5$ so A is invertible. If A is invertible, it cannot have free variables hence the solution for any consistent b must be unique.